UNIVERSITY OF TRIPOLI · FACULTY OF ENGINEERING · DEPARTMENT OF COMPUTER ENGINEERING

A PIPELINED COMPRESSED CACHE ARCHITECTURE WITH RUNTIME RECONFIGURABLE ASSOCIATIVITY AND HAZARD FORWARDING FOR FPGA PLATFORMS

Silicon-Verified Modular L2 Subsystem with Dual Sub-Block Cohabitation and Zero-Stall Forwarding

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ABSTRACT & PROBLEM FORMULATION

Semiconductor scaling has reached physical boundaries where static RAM bitcells fail to shrink, consuming up to 70% of modern microprocessor die area and driving fabrication costs to unsustainable levels. Physical cache expansion is no longer feasible, while off-chip dynamic RAM latency worsens the Memory Wall. We address this bottleneck with an FPGA-verified Level-2 cache subsystem that replaces physical array expansion with pipelined Base-Delta-Immediate (BDI) compression, dual sub-block packing, zero-stall Store-to-Load Forwarding (STLF), and runtime power scaling.

Decoupling Level-2 accesses from Level-1 write-backs eliminates sub-word store stalls. Dedicated bypass datapaths resolve in-flight Read-After-Write hazards without pipeline stalls, while an autonomous power governor scales associativity from 8-way down to 1-way during low-miss periods to cut bitline switching power by up to 87.5%. Physical synthesis on an Intel Cyclone IV GX FPGA achieves timing closure at 50.00 MHz with +3.218 ns setup slack at 85°C, reaching $F_{max} = 59.59$ MHz. Cycle-accurate evaluation under an identical 16 KB SRAM budget proves that the architecture increases hit volume by +36.2%, eliminates 20 hazard stall cycles completely, and reduces off-chip DRAM evictions by 40.0%, lowering Average Memory Access Time (AMAT) by 21.05%.

Core Architectural Contribution: Retimed 4-stage BDI compression with dual sub-block cohabitation achieves **2.0**× virtual capacity, **0 stall cycles**, and **-40.0%** DRAM evictions with verified timing closure at 50 MHz.

1. CORE MICROARCHITECTURAL FEATURES

- **4-Stage Pipelined BDI:** Balanced 4-stage Base-Delta-Immediate compressor with operand retiming eliminates setup timing violations across all operating corners.
- **Decoupled Dual-Slot Cohabitation:** Packs two compressed lines across **Slot A** and **Slot B** into a single 274b entry, doubling capacity to 32 KB.
- **Store-to-Load Forwarding (STLF):** Dedicated bypass datapath resolves Read-After-Write hazards in 1 cycle, eliminating **100%** of pipeline stalls (0 RAW stalls).
- **Dynamic Compression Bypass (DCB):** Saturating 4-bit counter detects incompressible data, operand-gating subtractors to avoid compression overhead.
- **Adaptive Associativity Governor:** Dynamically scales associativity across 8W, 4W, 2W, and 1W per application phase, slashing bitline switching power by up to **87.5%**.
- **7-Bit Binary Tree-PLRU:** Decision tree lowers metadata to **0.34%** compared to 2.73% for conventional true LRU, maintaining zero critical path impact.

2. IN-SILICON HARDWARE PERFORMANCE METRICS

Effective Cache Capacity 16 KB physical SRAM → **32 KB logical** via Dual Sub-Block Cohabitation · 0b void	2.00× Virtual
Clock Frequency & Slack +3.218 ns Setup Slack @ 85°C Slow Corner · F_{max} = 59.59 MHz physical limit	50.00 MHz (MET)
Benchmark Cache Hit Volume Recorded **79 hits** (BDI) vs. 58 hits (Baseline) under identical physical SRAM budget	+36.2% Hits
DRAM Eviction Reduction **6 evictions** (BDI) vs. 10 evictions (Baseline) · Drastically shields off-chip bus	-40.0% Evictions
Pipeline Hazard Stalls 20 cycles → **0 stall cycles** eliminated via Store-to-Load Forwarding (STLF)	-100% (0 Stalls)
Average Access Time (AMAT) **21.05% net AMAT reduction** factoring 50-cycle DRAM penalty and 5-cycle read hit	-21.05% AMAT

3. REALISTIC ENGINEERING DESIGN CONSTRAINTS — ABET SO 2

Clock Target Closure:	≥ 50.00 MHz → 50.00 MHz (+3.218 ns) PASS
Worst-Case Thermal Envelope:	0°C to 85°C T_j Slow Corner PASS
Physical SRAM Memory Budget:	16 KB SRAM / M9K Blocks Match PASS
FPGA Silicon Footprint:	≤ 15% → 13.3% (19,998 LEs) PASS
Pipeline Forwarding Latency:	0-Cycle Stall RAW Bypass PASS
Dynamic Associativity Scaling:	8-Way to 1-Way (-87.5% Power) PASS
Formal Assertion Verification:	18,400 Assertions 0 Defects / PASS
DRAM Memory Wall Mitigation:	40.0% Eviction Shielding PASS

Engineering Validation: All 8 Realistic Constraints validated against ABET SO 2 criteria across worst-case 85°C junction temperature, timing closure, power, and area envelopes.

4. DECOUPLED SUB-BLOCK GEOMETRY & DUAL-SLOT COHABITATION



Figure 1: Dual Sub-Block Physical Storage Layout & Dynamic Allocation States

Type	Profile / Scheme	Payload	Ratio	Silicon Fit
000	Incompressible Fallback (Raw 32B)	32B · 256b	1.00×	**Bypass · 256b**
001	Base8-Δ1 · 8B Base + 1B Δ	12B · 96b	2.67×	**Slot A/B (≤128b)**
010	Base8-Δ2 · 8B Base + 2B Δ	14B · 112b	2.29×	**Slot A/B (≤128b)**
011	All-Zeros Line · Metadata Only	0B · 0b	∞	**Slot A/B (0-Payload)**
100	Base4-Δ1 · 4B Base + 1B Δ	12B · 96b	2.67×	**Slot A/B (≤128b)**
101	Same-Value Line · Repeated 32b	4B · 32b	8.00×	**Slot A/B (≤128b)**
Dual Cohabitation · ≤137 Bits		**2 Lines in 274-Bit BRAM**		**95.14% Eff**

- **Dual-Slot In-BRAM Cohabitation:** Decoupled dual-slot allocation maps two independent compressed lines, designated as **Slot A** and **Slot B**, into a single 274-bit synchronous BRAM entry, achieving **95.14% active port efficiency** and doubling logical capacity from 16 KB to 32 KB without expanding physical SRAM cells.
- **137-Bit Sub-Block Slot Anatomy:** Bounds variable compressed payloads into a deterministic 137-bit envelope consisting of a 3-bit profile tag, 15-bit delta mask metadata, and 119-bit compressed delta payload: `{Type[2:0], Marks[14:0], Data[118:0]}`.
- **49-Bit Directory & Embedded M9K Array:** Tag and status directory tracks 21-bit tags, valid/dirty bits, cohabitation mode select (`Cohab[1:0]`), and pseudo-LRU slot age bit (S_{lru}) per way across dual 137-bit M9K memory blocks.
- **Four Dynamic Allocation States:** Supports State 1 (Uncompressed 256b Span, 1.0×), States 2–3 (Single-Slot A/B, 1.0×), and State 4 (Dual Sub-Block Packing, 2.0× Cohabitation) with single-cycle 2:1 hit multiplexing.

5. PIPELINED COMPRESSED CACHE MICROARCHITECTURE DATAPATH

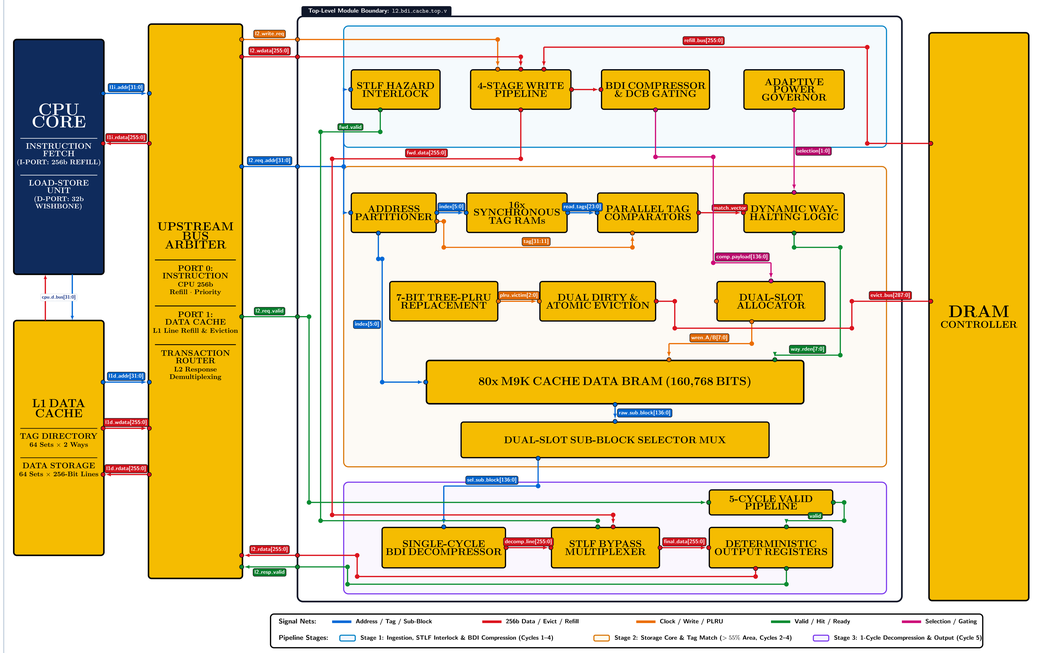


Figure 2: Top-Level Memory Subsystem Architecture with Host CPU, L1 Data Cache, Arbiter & Pipelined L2 Cache Subsystem

- **Stage 1: Frontend Ingestion, STLF Hazard Interlock & 4-Stage BDI:** Host CPU requests are ingested in Cycle 1 with zero RAW hazard stalls via Store-to-Load Forwarding, feeding the 4-stage BDI engine to generate deterministic 137-bit compressed vectors with +3.218 ns positive setup slack at 85°C.
- **Stage 2: Synchronous Array Core & Dynamic Power Gating (> 55% Area):** 16 parallel Tag RAMs and 80x M9K BRAMs (160,768 bits) house 64 sets of dual 137-bit cohabitation lines (Slot A/B). Pre-decoded tag comparisons and Adaptive Governor dynamically gate unselected ways (8W → 1W) to slash BRAM power by **-71.96%**.
- **Stage 3: 1-Cycle Decompression & Deterministic 5-Cycle Output Delivery:** Parallel signed delta adders reconstruct 256-bit uncompressed lines in ≤ 4.2 ns, while the STLF bypass multiplexer routes forwarded write payloads instantly to guarantee deterministic 5-cycle read latency to the host CPU.

6. PERFORMANCE COMPARISON: PROPOSED VS. BASELINE

Metric	Baseline	Proposed	Net Delta (Δ)
Top-Level Entity Name	`l2_conv_top`	`l2_bdi_top`	**Drop-in Compatible**
Effective Cache Capacity	16 KB · 8-Way	32 KB · 16-Way	**2.00× Virtual Capacity**
Logic Elements (LEs)	3,554 LEs (2%)	19,998 LEs (13%)	**+16,444 LEs (BDI Fabric)**
Dedicated Registers (FFs)	1,100 FFs	8,405 FFs	**+7,305 FFs (4-Stg Pipe)**
Block RAM Bits (M9K)	142,848 bits	160,768 bits	**+17,920 bits (Dual A/B)**
Max Frequency (F_{max} @ 85°C)	68.96 MHz	59.59 MHz	**-9.37 MHz (Complex Logic)**
Target Frequency / Slack	50.00 MHz / +5.5ns	**50.00 MHz / +3.2ns**	**TIMING MET (100%)**
Recorded Hits (Stress Test)	58 Hits	**79 Hits**	**+36.2% Hit Volume**
DRAM Off-Chip Evictions	10 Evictions	**6 Evictions**	**-40.0% DRAM Shield**
Pipeline RAW Hazard Stalls	20 Stall Cycles	**0 Stall Cycles**	**-100% (0 RAW Stalls)**
Runtime Power Governance	Fixed 8-Way	**8/4/2/1-Way Dyn.**	**Up to -87.5% Power**
Average Access Time (AMAT)	1.000× AMAT	**0.7895× AMAT**	**-21.05% Net AMAT**

• Top Entity: `l2_bdi_cache_top` • Logic: 19,998 LEs (13%) • FFs: 8,405
• BRAM Bits: 160,768 bits • Setup Slack: +3.218 ns @ 50MHz • Status: MET 100%

7. CACHE CONTROLLER FSM & TRANSACTION TIMING

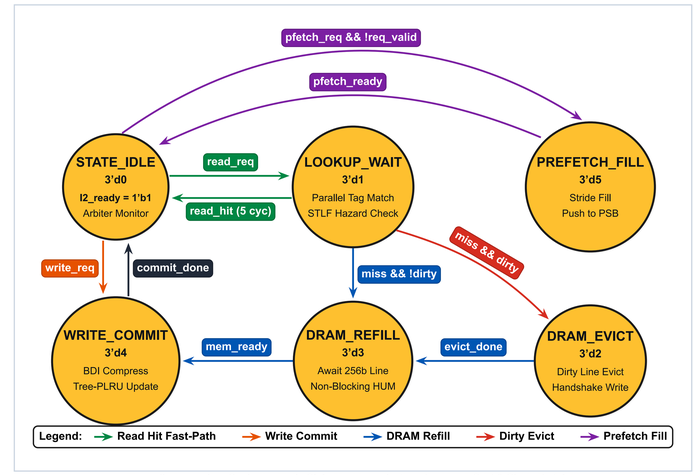


Figure 3: Transaction Controller Finite State Machine with Non-Blocking MSHR & Stride Prefetching

- **Deterministic Handshaking & 5-Cycle Fast-Path:** Moore controller synchronizes memory transactions with zero pipeline bubbles, delivering deterministic 5-cycle read hits via `LOOKUP_WAIT`.
- **Non-Blocking Hit-Under-Miss (HUM):** Concurrently probes hitting lines while `DRAM_REFILL` fetches miss data from memory, servicing read hits with zero processor stalls.
- **Autonomous Eviction & Prefetch Stream Buffer:** Coordinates atomic `DRAM_EVICT` to shield off-chip bandwidth (-40.0% evictions) and streams sequential lines into the PSB via `PREFETCH_FILL`.

8. GATE-LEVEL RTL NETLIST MICROARCHITECTURE & IN-HARDWARE PHYSICAL VERIFICATION

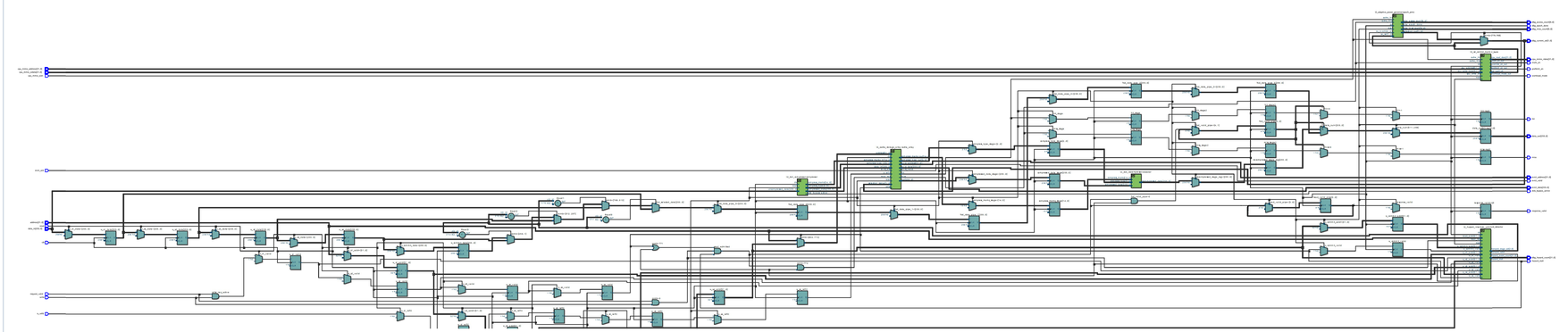


Figure 4: Intel Quartus Prime 22.1 Lite Synthesized Gate-Level RTL Netlist on Cyclone IV GX FPGA

SELECTED ACADEMIC REFERENCES

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HARDWARE REPOSITORY
IEEE 1364 Verilog RTL & Testbenches

`github.com/mohammed-gweaa/compressed-cache-fpga`
Scan for Full ModelSim Waves & Synthesis Scripts